SAMPLE

Cosmos Unveiled: The Cosmic Microwave Background

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Introduction

The cosmic microwave background (CMB, CMBR), or relic radiation, is microwave radiation that fills all space in the observable universe. With a standard optical telescope, the background space between stars and galaxies is almost completely dark. However, a sufficiently sensitive radio telescope detects a faint background glow that is almost uniform and is not associated with any star, galaxy, or other object. This glow is strongest in the microwave region of the electromagnetic spectrum.

Its energy density exceeds that of all the photons emitted by all the stars in the history of the universe. The accidental discovery of the CMB in 1964 by the American radio astronomers Arno Allan Penzias and Robert Woodrow Wilson was the culmination of work initiated in the 1940s. The CMB is the key experimental evidence of the Big Bang theory for the origin of the universe.

In the Big Bang cosmological models, during the earliest periods, the universe was filled with an opaque fog of dense, hot plasma of sub-atomic particles. As the universe expanded, this plasma cooled to the point where protons and electrons combined to form neutral atoms of mostly hydrogen. Unlike the plasma, these atoms could not scatter thermal radiation by Thomson scattering, and so the universe became transparent.

Known as the recombination epoch, this decoupling event released photons to travel freely through space.

However, the photons have grown less energetic due to the cosmological redshift associated with the expansion of the universe. The surface of last scattering refers to a shell at the right distance in space so photons are now received that were originally emitted at the time of decoupling. The CMB is very smooth and uniform, but maps by sensitive detectors detect small but important temperature variations.

Ground and space-based experiments such as COBE, WMAP and Planck have been used to measure these temperature inhomogeneities.

The monopole term, l = 0, is the constant isotropic mean temperature of the CMB, Tgamma = 2.7255+/-0.0006 K with one standard deviation confidence. This term must be measured with absolute temperature devices, such as the FIRAS instrument on the COBE satellite. The dipole term, at index one:

Measurements of the density of quasars based on Wide-field Infrared Survey Explorer data finds a dipole significantly different from the one extracted from the CMB anisotropy. This difference is in conflict with the cosmological principle.

Robert Dicke, James Peebles, P. G. Roll, and D. T. Wilkinson interpret this radiation as a signature of the Big Bang. 1966 - Rainer K. Sachs and Arthur M.

Depending on the moment of injection, this causes a distortion, which can be characterized using so-called

CMB spectral distortions therefore provide a powerful probe of early-universe physics and even deliver crude estimates for the epoch at which the injection occurred.

Richer distortion signals, going beyond the classical

The cosmological recombination radiation (CRR) is a prime example within

CDM that is created by photon injection from the recombining hydrogen and helium plasma around redshifts of

After COBE/FIRAS provided stringent limits on the CMB spectrum, essentially ruling out distortions at the level

Robert Dicke predicts a microwave background radiation temperature of 20 K.

Tr = 2.725 K x (1 + z)

or stated plainly: the relic temperature scales with the stretch factor itself

A framing observation sets the afterglow's paradox before the chapters begin. The cosmic microwave background is simultaneously the most perfectly smooth thing ever measured and the most informative, a glow the same in every direction to five decimal places whose remaining flicker, read carefully, gives the age of the universe, its geometry, its contents, and the recipe of the galaxies; this book is about the light left over from the beginning, and the chapters that follow trace its arc from an unexplained hiss in a New Jersey antenna to the foundation of the modern cosmology, the standard model of the universe's history, written in the oldest light there is.

Chapter 01: The Discovery of The Cosmic Microwave Background

The discovery of the cosmic microwave background is one of the great accidents in the history of science, a story worth telling in full. In the 1940s, the physicists who took the expanding universe seriously worked out what a hot, dense origin implied: the early universe must have been filled with radiation, and that radiation, stretched and cooled by the expansion itself, should still exist, a faint glow filling all of space at a temperature of a few kelvin. The prediction was made, largely ignored, and then made again, and by the early 1960s a group at Princeton had begun building the apparatus to detect the leftover glow, expecting a temperature of a few degrees above absolute zero.

Forty kilometers away, two radio engineers were fighting a noise they could not explain. Arno Penzias and Robert Wilson, working at Bell Telephone Laboratories, had built a sensitive horn antenna for satellite communication, and no matter how they pointed it, the instrument read a steady hiss, corresponding to a temperature of a few kelvin, that would not go away. They checked everything: the electronics, the cables, the sky, the ground. They cleaned pigeon droppings out of the horn and considered the birds themselves a possible source. The noise remained, constant across the sky and through the seasons, arriving equally from every direction, which was the signature of something that was not on the Earth and not in the galaxy but everywhere.

The connection was made through a telephone call and a friend of a friend. The Princeton group, having heard of the Bell Labs mystery, recognized their predicted signal in the engineers' stubborn noise, and in 1965 the two teams published companion papers: one announcing the measurement, the other the interpretation. The universe had a temperature. The event is the origin story of modern cosmology, and its two lessons are worth noting: that the most important signal in the history of astronomy was found by people who were not looking for it, and that it was understood only because someone else had done the theory in advance; the serendipity was real, but it was prepared serendipity, the meeting of a stubborn measurement with a ready explanation.

The confirmation that followed settled the case. The glow's spectrum was measured over decades, and it matches the blackbody curve, the smooth thermal spectrum of a system in equilibrium, to extraordinary precision, a fact the alternative theories of the day could not reproduce; and the discovery's Nobel Prize arrived a decade later, while the Princeton apparatus that was built to find the signal first is remembered as the second instrument to find it. The reader should carry the discovery chapter's moral forward into the rest of the book, that the afterglow was predicted, found, explained, and only then believed, which is the proper order, and that every fact in the chapters that follow descends from a hiss that two engineers once spent a year trying to make go away.